

# CanonicalForms

## Abstract

We formulate, in the language of set theory, the concepts of invariant and canonical form for a binary relation  $\sim$  on a set  $X$ , and study them mainly when  $\sim$  is an equivalence relation. A map  $f: X \rightarrow Y$  is called an invariant of  $\sim$  if  $x \sim y \implies f(x) = f(y)$ , and a complete invariant if equality of images also implies  $x \sim y$ . A canonical form is an invariant  $C: X \rightarrow X$  satisfying  $x \sim C(x)$  for all  $x \in X$ ; its image  $C(X)$  is called the skeleton of  $\sim$ , and, when  $\sim$  is an equivalence relation, also the complete system of representatives of  $\sim$ . We characterise canonical forms via retraction–section pairs, showing that  $C$  is a canonical form if and only if there exists a section  $s$  of the quotient map  $q: X \rightarrow X/\sim$  such that  $C = s \circ q$ . An analogous characterisation holds for surjective complete invariants. We prove that every complete invariant  $I: X \rightarrow Y$  factors uniquely through the surjective complete invariant  $I': X \rightarrow I(X)$ , and define  $s' \circ I'$  as the canonical form of  $\sim$  determined by the complete invariant  $I$ , whenever a section  $s'$  of  $I'$  exists. Assuming the axiom of choice, we show that the canonical form determined by any complete invariant exists. Propositions that assume the axiom of choice are marked with <sup>AC</sup>.

**Definition 1** Let  $X, Y$  be sets. Let  $f: X \rightarrow Y$  be a map and let  $\sim$  be a binary relation on  $X$ . We say that  $f$  is an **invariant** of  $\sim$  if the following condition holds.

$$\forall x, y \in X, (x \sim y \implies f(x) = f(y))$$

We further say that  $f$  is a **complete invariant** of  $\sim$  if the following condition holds.

$$\forall x, y \in X, (x \sim y \iff f(x) = f(y))$$

□

**Example 1** Let  $X$  be a set. Let  $\sim$  be an equivalence relation on  $X$  and let  $q: X \rightarrow X/\sim$  be the quotient map. Then  $q$  is a complete invariant of  $\sim$ . □

**Definition 2** Let  $X$  be a set. Let  $C: X \rightarrow X$  be an invariant of a binary relation  $\sim$  on  $X$ . We say that  $C$  is a **canonical form** of  $\sim$  if the following condition holds.

$$\forall x \in X, x \sim C(x)$$

We call the image  $C(X)$  of  $X$  under  $C$  the **skeleton** of  $\sim$ . When  $\sim$  is an equivalence relation, we also call  $C(X)$  the **complete system of representatives** of  $\sim$ .  $\square$

**Proposition 1** Let  $X$  be a set. Let  $C: X \rightarrow X$  be a canonical form of a binary relation  $\sim$  on  $X$ . Then  $C = C \circ C$ .  $\square$

**Proof** Let  $x \in X$ . Since  $x \sim C(x)$ , we have  $C(x) = C(C(x))$ . Therefore  $C = C \circ C$ .  $\blacksquare$

**Definition 3** Let  $X, Y$  be sets and let  $r: X \rightarrow Y$  and  $s: Y \rightarrow X$  be maps. We call the pair  $(r, s)$  a **retraction-section pair** if the following condition holds. We call  $r$  the **retraction** of  $s$ , and  $s$  the **section** of  $r$ .

$$r \circ s = \text{id}_Y$$

$\square$

**Lemma 1** Let  $X$  be a set and let  $q: X \rightarrow X/\sim$  be the quotient map of an equivalence relation  $\sim$  on  $X$ . For a map  $C: X \rightarrow X$ , the following conditions (1) and (2) are equivalent.

- (1)  $C$  is a canonical form of  $\sim$ .
- (2) There exists a section  $s: X/\sim \rightarrow X$  of  $q: X \rightarrow X/\sim$  such that  $C = s \circ q$ .

$\square$

**Proof** (1)  $\Rightarrow$  (2): Assume  $C$  is a canonical form of  $\sim$ . For each  $y \in X/\sim$ , since  $q$  is surjective there exists  $x \in X$  such that  $y = q(x)$ . Set  $s(y) = C(x)$ . For any  $x' \in X$  with  $q(x) = q(x')$ , we have  $x \sim x'$ , so  $C(x) = C(x')$ . Thus  $s: X/\sim \rightarrow X$  is well-defined. For any  $x \in X$ ,  $C(x) = s(q(x))$ , so  $C = s \circ q$ . For any  $x \in X$ ,  $(q \circ s)(q(x)) = q(s(q(x))) = q(C(x))$ , and since  $x \sim C(x)$  we have  $q(C(x)) = q(x)$ . Therefore  $q \circ s = \text{id}_{X/\sim}$ .

(2)  $\Rightarrow$  (1): Assume there exists a section  $s: X/\sim \rightarrow X$  of  $q: X \rightarrow X/\sim$  such that  $C = s \circ q$ . For  $x, x' \in X$ , if  $x \sim x'$ , then  $C(x) = (s \circ q)(x) = s(q(x)) = s(q(x')) = (s \circ q)(x') = C(x')$ , so  $C$  is an invariant of  $\sim$ . For any  $x \in X$ ,  $q(x) = \text{id}_{X/\sim}(q(x)) = (q \circ s)(q(x)) = q((s \circ q)(x)) = q(C(x))$ , so  $x \sim C(x)$ . Therefore  $C$  is a canonical form of  $\sim$ .  $\blacksquare$

**Proposition 2** Let  $X, Y$  be sets and let  $I: X \rightarrow Y$  be a surjective complete invariant of an equivalence relation  $\sim$  on  $X$ . For a map  $C: X \rightarrow X$ , the following conditions (1) and (2) are equivalent.

- (1)  $C$  is a canonical form of  $\sim$ .
- (2) There exists a section  $s': Y \rightarrow X$  of  $I: X \rightarrow Y$  such that  $C = s' \circ I$ .  $\square$

**Proof** By Lemma 1, it suffices to show that the existence of a section  $s: X/\sim \rightarrow X$  of  $q: X \rightarrow X/\sim$  with  $C = s \circ q$  is equivalent to (2). Define  $\bar{I}: X/\sim \rightarrow Y$  by  $\bar{I}(q(x)) = I(x)$  for all  $x \in X$ . When  $x \sim x'$ , since  $I$  is an invariant we have  $I(x) = I(x')$ , so  $\bar{I}$  is well-defined. For any  $x, x' \in X$ , if  $I(x) = I(x')$  then, since  $I$  is a complete invariant,  $x \sim x'$ , hence  $q(x) = q(x')$ , so  $\bar{I}$  is injective. For any  $y \in Y$ , since  $I: X \rightarrow Y$  is surjective there exists  $x \in X$  with  $I(x) = y$ , giving  $\bar{I}(q(x)) = y$ . Therefore  $\bar{I}$  is bijective and hence invertible.

Suppose there exists a section  $s$  of  $q$  with  $C = s \circ q$ . Set  $s' = s \circ \bar{I}^{-1}$ . Since  $q = \bar{I}^{-1} \circ I$ , we have  $C = s \circ \bar{I}^{-1} \circ I = s' \circ I$ . Moreover,  $I \circ s' = (\bar{I} \circ q) \circ (s \circ \bar{I}^{-1}) = \bar{I} \circ (q \circ s) \circ \bar{I}^{-1} = \bar{I} \circ \text{id}_{X/\sim} \circ \bar{I}^{-1} = \text{id}_Y$ , so  $s'$  is a section of  $I$ .

Conversely, suppose (2) holds. Set  $s = s' \circ \bar{I}$ . Then  $s \circ q = s' \circ \bar{I} \circ q = s' \circ I = C$ . Since  $I \circ s' = \text{id}_Y$ , we have  $I \circ s' \circ I = I$ , and therefore  $I \circ C = I$ . Since  $I = \bar{I} \circ q$ , it follows that  $\bar{I} \circ q \circ C = \bar{I} \circ q$ , and by injectivity of  $\bar{I}$  we obtain  $q \circ C = q$ . For any  $y \in Y$  there exists  $x \in X$  such that  $y = q(x)$ . Then

$$(q \circ s)(y) = (q \circ s' \circ \bar{I})(y) = (q \circ s' \circ \bar{I} \circ q)(x) = (q \circ s' \circ I)(x) = (q \circ C)(x) = q(x) = y,$$

so  $q \circ s = \text{id}_{X/\sim}$ . ■

**Proposition 3** Let  $X, Y$  be sets and let  $I: X \rightarrow Y$  be a complete invariant of an equivalence relation  $\sim$  on  $X$ . The unique map  $I': X \rightarrow I(X)$  satisfying  $I = i \circ I'$  with respect to the inclusion  $i: I(X) \rightarrow Y$  is surjective and is a complete invariant of  $\sim$ . □

**Proof** Define  $I': X \rightarrow I(X)$  by  $I'(x) = I(x)$  for all  $x \in X$ . Then  $i(I'(x)) = I(x)$  for all  $x \in X$ , so  $I = i \circ I'$ ; this shows that such a map exists. Since  $I'(X) = I(X)$ ,  $I'$  is surjective. For any  $x, x' \in X$ ,  $x \sim x' \iff I(x) = I(x') \iff I'(x) = I'(x')$ , so  $I'$  is a complete invariant of  $\sim$ . If  $I'': X \rightarrow I(X)$  also satisfies  $I = i \circ I''$ , then  $i(I'(x)) = I(x) = i(I''(x))$  for all  $x \in X$ , and since  $i$  is injective,  $I'(x) = I''(x)$ . Therefore  $I'' = I'$ , so  $I'$  is unique. ■

**Definition 4** Let  $X, Y$  be sets and let  $I: X \rightarrow Y$  be a complete invariant of an equivalence relation  $\sim$  on  $X$ . When a section  $s': I(X) \rightarrow X$  of the unique map  $I': X \rightarrow I(X)$  satisfying  $I = i \circ I'$  with respect to the inclusion  $i: I(X) \rightarrow Y$  exists, we call the canonical form  $s' \circ I'$  of  $\sim$  the **canonical form of  $\sim$  determined by the complete invariant  $I$** . □

**Lemma 2**<sup>AC</sup> Let  $X, Y$  be sets and let  $f: X \rightarrow Y$  be a surjection. Then there exists a map  $g: Y \rightarrow X$  such that  $f \circ g = \text{id}_Y$ . □

**Proof** Since  $f: X \rightarrow Y$  is surjective,  $f^{-1}(y) \neq \emptyset$  for every  $y \in Y$ . Therefore, by the axiom of choice,  $\prod_{y \in Y} f^{-1}(y) \neq \emptyset$ , so there exists  $g \in \prod_{y \in Y} f^{-1}(y)$ . This map  $g: Y \rightarrow X$  satisfies  $f \circ g = \text{id}_Y$ . ■

**Proposition 4**<sup>AC</sup> Let  $X, Y$  be sets and let  $I: X \rightarrow Y$  be a complete invariant of an equivalence relation  $\sim$  on  $X$ . Then the canonical form of  $\sim$  determined by the complete invariant  $I$  exists. □

**Proof** Let  $q: X \rightarrow X/\sim$  be the quotient map. Since  $q$  is surjective, by Lemma 2 there exists a map  $s: X/\sim \rightarrow X$  satisfying  $q \circ s = \text{id}_{X/\sim}$ . By Lemma 1,  $s \circ q$  is a canonical form of  $\sim$ , and by Proposition 2, taking  $I'$  to be the surjection determined by  $I$ , there exists a section  $s': I(X) \rightarrow X$  of  $I'$  such that  $s \circ q = s' \circ I'$ . Then  $s' \circ I'$  is the canonical form of  $\sim$  determined by the complete invariant  $I$ . ■